

Josué García

Full Stack Developer | Java, Spring Boot, React & PostgreSQL
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Professional Summary

Full Stack Developer who builds complete systems end to end: REST APIs, web interfaces, databases, microservices and real business workflows, primarily with Java and Spring Boot, Angular and React over PostgreSQL. Hands-on experience with distributed architectures, containerization with Docker, CI/CD pipelines and automated testing, along with cloud deployment on AWS and Azure, backed by a solid foundation in databases and infrastructure.

Technical Skills

Languages: Java, TypeScript, JavaScript, Python, PHP, SQL, HTML, CSS

Frameworks: Spring Boot, Spring Cloud, Spring Security, Angular, React, NestJS, Node.js, Laravel, Tailwind CSS

Databases: PostgreSQL, MySQL, Oracle, Redis, JPA / Hibernate, Citus, PostGIS

Cloud & DevOps: Docker, Kubernetes, Helm, Jenkins, GitHub Actions, CI/CD, Nginx, Certbot, Cloudflare, AWS, Azure

Testing & Quality: JUnit 5, Mockito, JaCoCo, SonarQube, Gatling, JMeter, OWASP ZAP, Selenium, Playwright, Postman

Architecture & Integration: REST APIs, Microservices, gRPC, Eureka, API Gateway, WebSocket, OpenAPI / Swagger, JWT

AI & Computer Vision: YOLOv11, U-Net, CNN, PyTorch, TensorFlow, OpenCV

Tools & Methods: Git, Linux, Maven, Odoo, Scrum, Agile

Technical Projects

PetSafe — Veterinary Clinic Management

Mar 2026 – May 2026

NestJS, Angular, TypeScript, PostgreSQL, WebSockets, Docker, CI/CD

- Developed several full-stack modules (Angular + NestJS): appointment scheduling, pets and owners, adoptions, settings and reporting, plus a dashboard with operational metrics (daily and weekly appointments).
- Implemented the real-time attention queue with WebSockets, JWT authentication and API documentation with OpenAPI.
- Built the system test suite with Playwright and the CI/CD pipeline to a VPS.
- Contributed to the production deployment: Nginx configuration, SSL certificates with Certbot, domain management with Cloudflare and operation on a VPS.

Hospital Management System

Sep 2025 – Oct 2025

Java, Spring Boot, Spring Cloud, Microservices, PostgreSQL, Azure, React

- Developed the authentication service (JWT) and the API Gateway, with service discovery (Eureka) and resilience via Circuit Breaker, and built the patients and reporting modules.
- Set up distributed PostgreSQL with Citus (sharded by medical center) and deployed the microservices on Azure with Docker; also contributed the patient management and login frontend in React.

Event Booking System (Ticketing)

Feb 2026

Java 21, Spring Boot 3, PostgreSQL, Redis, Gatling, Docker

- Single-handedly developed a backend for ticket booking in high-demand events, focused on concurrency control and data consistency.
- Validated the system with Gatling load tests over 1,000 simultaneous bookings, ensuring 100% consistency and no duplicates.
- Implemented distributed locking with Redis for seat contention (–67% median latency) and optimized the transactional scope (+25% throughput).

Publication Management and Moderation System

Oct 2025 – Dec 2025

Java, Spring Boot, React, PostgreSQL, PostGIS, Docker, CI/CD

- Developed the incident and appeal moderation module (backend and frontend), with its business logic and email handling, plus proximity-based product discovery via geospatial queries with PostGIS.
- Implemented JWT security and a quality and security pipeline (SonarQube, JaCoCo, JMeter, OWASP ZAP) with E2E tests (Selenium, Playwright) integrated in Jenkins.

Distributed Fuel and Fleet Management System

Sep 2025 – Dec 2025

NestJS, gRPC, PostgreSQL, Kubernetes, Docker

- Developed the routes (with map) and trips modules, including estimated-vs-actual fuel calculation, per-trip efficiency metrics and their reporting.
- Implemented gRPC communication between microservices (NestJS) behind an API Gateway.

Iron Zone — Gym Management on Odoo

Apr 2026 – Jun 2026

Odoo 18, Python, PostgreSQL, Docker, XML

- Developed the subscriptions module (memberships and recurring billing) and the role and permission management, with administrative views and role-based access control.
- Applied fixes to the website and testing, and deployed the solution with Docker and continuous integration, configuring Nginx and the domain SSL certificates with Certbot.

Savings and Financial Goals Simulator
Laravel, React, JWT, MySQL

Apr 2025

- Built the frontend (goals, contributions, progress panel, suggestions and reports) and the corresponding business logic in the backend with JWT authentication, delivered end to end in a competition sprint.

Asset Maintenance Management System
Laravel, React, JWT, MySQL

Dec 2024 – Jan 2025

- Developed asset management and their responsible staff, along with the replaced-components and suppliers modules, in a full-stack platform built as a team under Scrum.

Creative Portfolio Platform
PHP, MySQL, JavaScript

Mar 2024 – Jun 2024

- Developed the portfolios and publications module with image uploads (JavaScript) and per-user visibility control, plus the comment system with nested replies, without frameworks.

License Plate Recognition and Vehicle Monitoring
Python, YOLOv11, U-Net, CNN, PyTorch, FastAPI, PostgreSQL

May 2026 – Jun 2026

- Developed license plate detection with YOLOv11 and strengthened segmentation (U-Net) and OCR with post-processing that discards false characters (screws, spurious regions).
- Implemented speed estimation through virtual lines and the generation of a suggested, interpretable penalty with a Mamdani fuzzy engine, integrating the flow into a web monitoring application.
- Evaluated on 74 real vehicles: 94.59% accuracy reading the full plate, with a detector at 96.13% mAP50.

Maternal Risk Assessment with a Fuzzy-Genetic System
Python, FastAPI, React, Fuzzy Logic, Genetic Algorithms

Apr 2026 – Jun 2026

- Designed and implemented the genetic algorithm (Michigan approach) that learns the fuzzy rule base, reducing a space of 8,640 candidate rules to a compact base of 159 rules.
- Built the comparative model-evaluation flow (genetic algorithm vs. RIPPER vs. PRISM) over 20 runs with mean and standard deviation in accuracy and balanced accuracy (GA 0.4937 / 0.5033; final 0.5664 / 0.5629).
- Defined part of the fuzzy logic (low/mid/high thresholds derived from class-distribution proportions) and an explainable interface that shows the rules fired for each prediction.

Digital Image Processing Pipeline
Python, NumPy, OpenCV

May 2026 – Jun 2026

- Implemented most of the pipeline from scratch: grayscale conversion, normalization, smoothing, sharpening, gradient (edge detection) and character-box detection, with hand-written convolution and interactive per-stage visualization.

Nature Reserve Manager
Java, Data Structures, OOP

Mar 2024 – Apr 2024

- Implemented the iterator of the generic linked list and part of the criteria-based search (Criteria pattern), along with domain modules such as incidents, in Java.

Education

B.Eng. in Software Engineering — Universidad Técnica de Ambato

2022 – Present

Certifications

Spring Framework 6 and Spring Boot 3 (Udemy) | Relational Databases – Intermediate (Udemy)

Languages

Spanish: native | English: intermediate